



A ROCKWELL AUTOMATION COMPANY

Digital Transformation Service Delivery for Enterprises & Industries

Knowledge Lens & Kalypso deliver digital transformation for Rockwell Automation.

We help our clients discover, create, make and sell better products with digital.

Rockwell Automation's
Digital Services arm

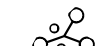
Recognized Digital Leader

Gartner

THE CHANNEL CO.
CRN

ARC
Advisory Group

 **IDC**
ANALYZE THE FUTURE


IOT ANALYTICS
MARKET INSIGHTS FOR THE INTERNET OF THINGS

Knowledge Lens
A ROCKWELL AUTOMATION COMPANY

KALYPSO
A ROCKWELL AUTOMATION BUSINESS

COMPREHENSIVE
CAPABILITIES

INDUSTRY
FOCUS



Chemicals &
Life Sciences



Metals, Cement,
Heavy Industries



Industrial
High Tech

Consulting

Digital Thread Strategy
Product & Portfolio Management (PPM)
Operations Performance Improvement
Supply Chain Optimization
Sustainability
Organizational Change Management

Data Science & AI

AI-enabled smart factory & connected worker (Industry 5.0)
Sustainability & Net ZERO
Product Lifecycle Intelligence
Autonomous Manufacturing
Model Predictive Control
Predictive Service & Maintenance

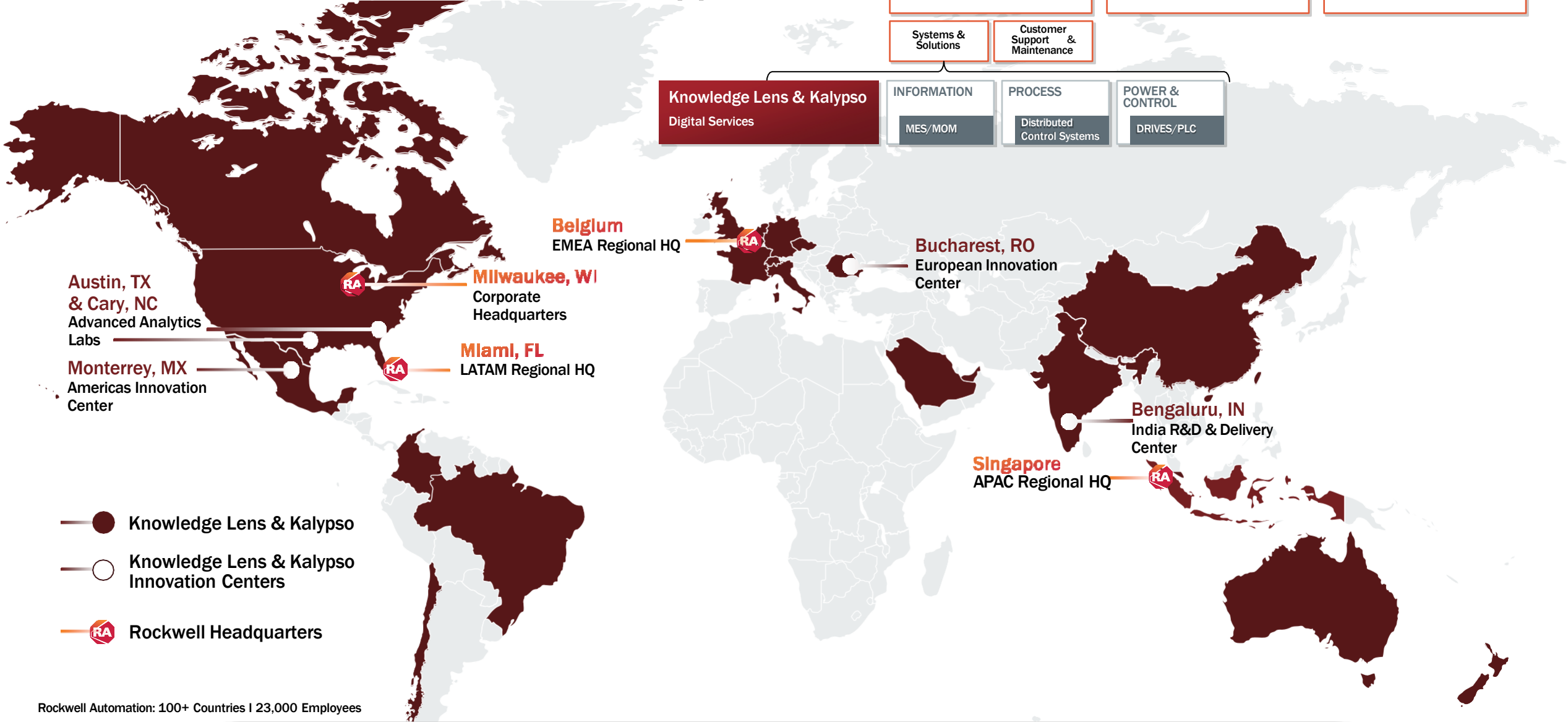
Technology

Digital Twin
Product & Data Lifecycle Management
Supply Chain Management (SCM)
Enterprise Resource Planning (ERP)
End-to-end enterprise & manufacturing data lakes
Extended Reality (AR/VR/MR)

Managed Services

Data Lake support
Cloud infrastructure administration & Hosting
Validation & documentation support
Application Management
Continuous Improvement
Solution Services

Global operating model combines world-class resources with localized support



Our Solutions **resonate cross-industry; validated** across industries



Pharma & Chemical



Textile



Refinery & Energy



Metals & Minerals



Cement



CPG



Our core offerings are aligned with a digital enterprise's most critical engineering needs

Enterprise Analytics & AI



- End-to-end enterprise & manufacturing data lakes
- Blockchain
- Data Science, AI, & ML

Cloud Engineering



- Multi-cloud infrastructure setup
- DevOps/MLOps + microservices & containerization
- Cloud migration & app modernization

Industrial IoT & Sustainability



- AI-enabled smart factory & connected worker (Industry 5.0)
- IT/OT integration across edge & cloud
- Environment monitoring, reporting, & analytics

Application Development



- Custom applications
- Digital product engineering
- Mobile apps (iOS & Android)

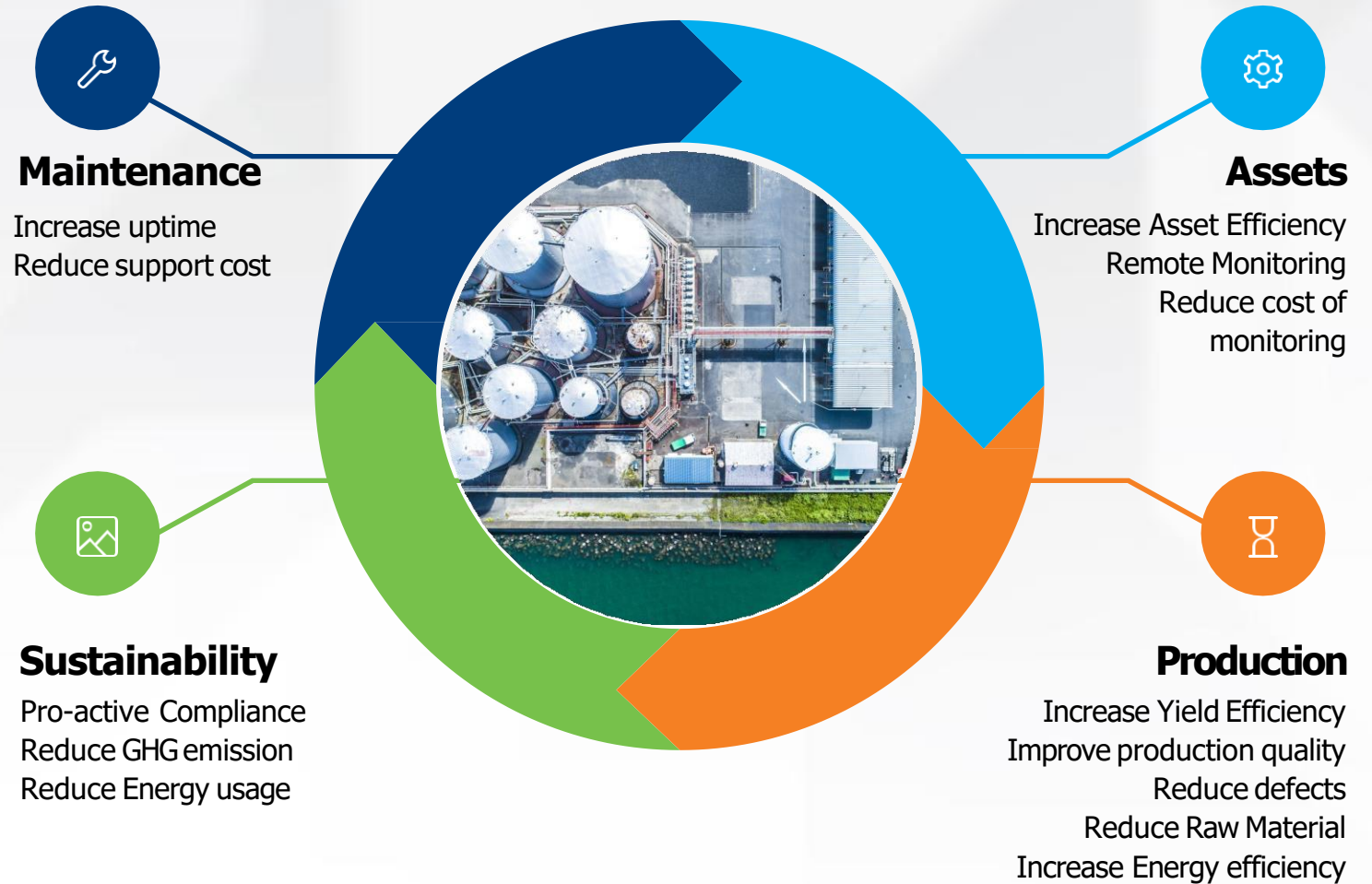
Application Maintenance



- Data lake support
- Cloud infrastructure administration
- Validation & documentation support

Our Vision

Enable Sustainable
Business Outcomes using
Intelligent Industrial Apps,
across the manufacturing
lifecycle



Empowering Industries at every Level with our Solutions

Our apps cater to
different persona

From **plant floor to boardroom**

Validated across
personas



Our customer success stories reflect proven solutions deployed at scale using cutting-edge digital technologies (1/2)

Enterprise Data Lake

Ground-up architecture design, build, & support of a GMP data lake across sales, R&D, supply chain, manufacturing, quality

Customer – Top-15 biopharmaceutical company

Deployment – Cloud

Scale – 75+ data sources, 8,000+ active users

In Production – 6 years

Manufacturing Data Lake

Smart factory & connected worker solution based on contextualized L1, L2, L3, & L4 shop floor data, including intelligent-EBR & yield optimizer apps

Customer – Global chemicals company

Deployment – Cloud

Scale – 10 manufacturing plants

In Production – 2 years

Enterprise Cloud Migration

Migration of data & workloads from legacy on-prem applications to the cloud

Customer – Top-5 biopharmaceutical company

Deployment – Cloud

Scale – 150+ data sources

In Production – 4 years

Environmental Pollution Monitoring

End-to-end IIoT solution for real-time monitoring, reporting, & analytics of emissions & effluents

Customer – Cross-industry & government (top-5 GDP)

Deployment – Multi-tenant cloud & mobile app

Scale – 150+ cities & 2,500 manufacturing plants

In Production – 8 years

AI for Hi-Tech Manufacturing

Real-time predictive control for circuit board manufacturing nozzles on the shop-floor

Customer – Top-3 industrial automation company

Deployment – Cloud, Edge (containers), mobile app

Scale – 6 manufacturing plants

In Production – 3 years

Vision-Based AI for Product Quality

Real-time in-line monitoring of food product quality (color, dimensions, impurities, etc.)

Customer – \$1B consumer goods company

Deployment – Edge

Scale – 3 sites

In Production – 2 years

Our customer success stories reflect proven solutions deployed at scale using cutting-edge digital technologies (2/2)

Vision-Based AI for Flare Monitoring

Real-time video monitoring & analytics using AI for the detection of combustion in flare stacks.

Customer – Top-3 oil company
Deployment – Edge (containers)
Scale – 2 sites
In Production – 3 years

Digital Twin for Chiller Optimization

Real-time performance monitoring & recommendations to optimize chiller sequencing & loading.

Customer – \$50B manufacturing conglomerate
Deployment – Cloud (Azure), Edge
Scale – 20+ assets
In Production – 2 years

Remote Equipment Monitoring

Remote tracking of coolers & refrigerators across retailers, including parts & service management.

Customer – Top-10 consumer goods company
Deployment – Cloud (Azure)
Scale – 50,000+ assets
In Production – 3 years

Solar Panel Monitoring

Remote monitoring of solar panels, inverters, etc., including ticketing & service management

Customer – \$1B solar renewable energy company
Deployment – Cloud (Azure)
Scale – 300+ sites
In Production – 2 years

AI for Industrial Health & Safety

Real-time AI-based detection of equipment leaks, PPE compliance, intrusions, etc.

Customer – \$45B oil company
Deployment – Cloud (Azure)
Scale – 1 site
In Production – 1 years

AI-Enabled Sales Force

AI-based real-time recommendations for pharma sales representatives in the field

Customer – \$1B consumer goods company
Deployment – Cloud
Scale – 3,000+ sales reps
In Production – 3 years

Our Differentiation

Remote Monitoring



API's & No Code App Tools

AI driven Recommendation & Insights

Machine Data

Asset Data
Tags/Parameters
DCS

Asset Analytics



Yield Optimization



Energy Manager



Sustainability



Industrial Knowledge Graph

Edge to Cloud Connectivity

Operator/Human Data

Production Data
Quality Testing Data
Maintenance SOP/Checklist

Knowledge Lens
A ROCKWELL AUTOMATION COMPANY

- We believe in contextualizing human and machine data to empower business outcomes
- We seamlessly discover knowledge of industrial operations from machine events and human actions to create the Industrial Knowledge Base

- We leverage the Industrial Knowledge Graph to deliver recommendations and insights to front line workers
- We believe in concept of Industrial App Suite to cater to business outcomes related to all phases of industrial life cycle

All apps offered as SaaS or On-prem; comes with complete edge to cloud components



Sustainability

A scalable Cloud DAHS for Environment Compliance and Forecasting along with Net Zero Carbon Accounting



Energy Manager

Cloud based Energy Management and Optimization Application for Energy cost savings and Sustainability goals



Asset Analytics

Digital Twin model driven asset analytics app for real time efficiency recommendations and predictive maintenance of critical assets



Yield Optimization

AI-based app for Continuous/Batch process yield and raw material optimization with real time recommendation



Remote Monitoring

Cloud managed SaaS offering that remotely configure edge to communicate with assets and enables data monitoring and analytics in cloud

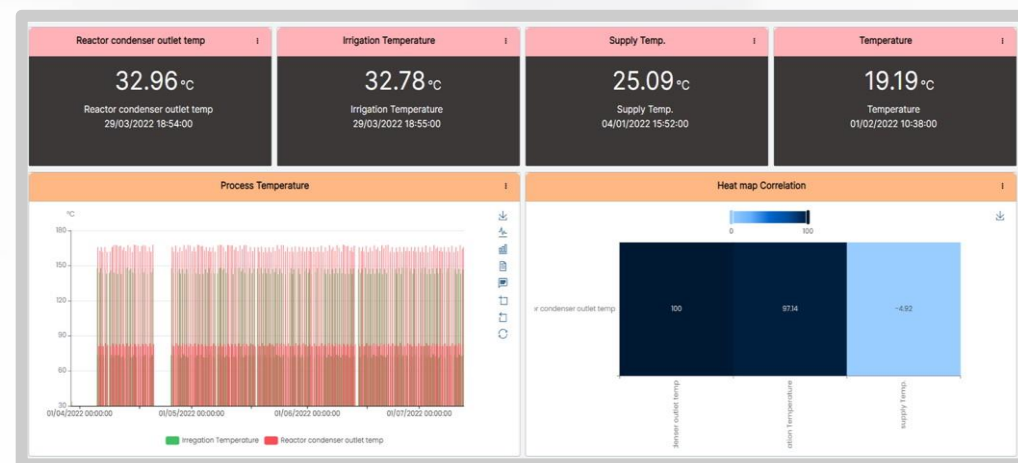
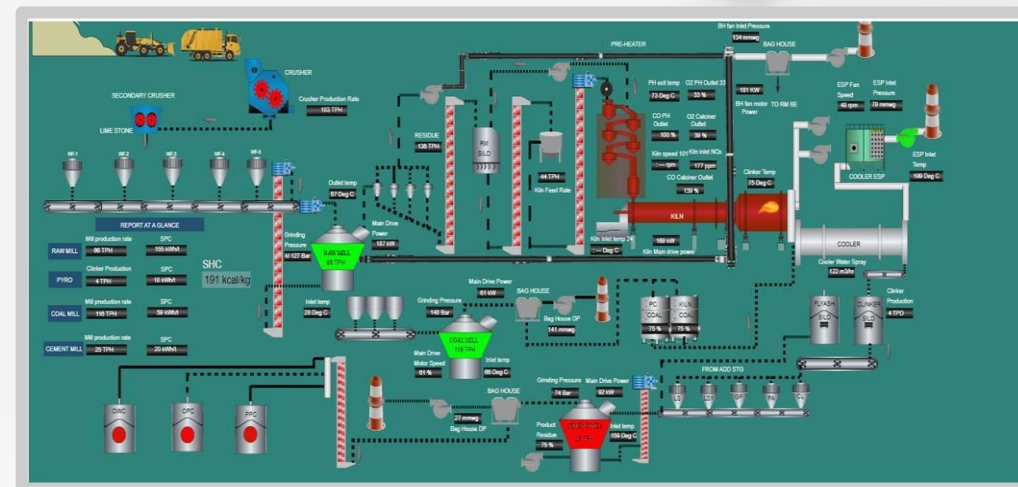
Remote Monitoring

Solution Overview

Remote Monitoring is an app that helps industries kick start their digital transformation to Industry 5.0. It enables edge to cloud data connectivity of industrial assets and helps bring the data to cloud for remote monitoring and analytics

Key Features

- ✓ **Asset Management:** Highly customizable and easily configurable module to define the Asset Hierarchy, define Asset Models and view data for all the Assets using the Asset Explorer
- ✓ **Edge Management:** Highly scalable Edge Manager that can facilitate protocol configuration, enablement of data acquisition, centralized health monitoring and OTA upgrades for edge gateways
- ✓ **Parameter Monitoring with KPI Hierarchy:** Out of the box web/mobile app that provides a flexible configuration tool to define and organize multiple hierarchies of KPIs, their corresponding alerting rules, workflows and escalation matrix for deviations
- ✓ **Digital Twin:** Real time monitoring of any process thru the Digital Twin on the Web/Mobile interfaces with flexibility to highlight anomalies and color codes on the Digital Twin interface
- ✓ **Self-service Analytics:** The data made available on the cloud is completely open and enabled for self-service analytics thru the Data Explorer as well 50+ self-service visualization widgets starting from Line/Bar charts to advanced widgets for SPC analytics



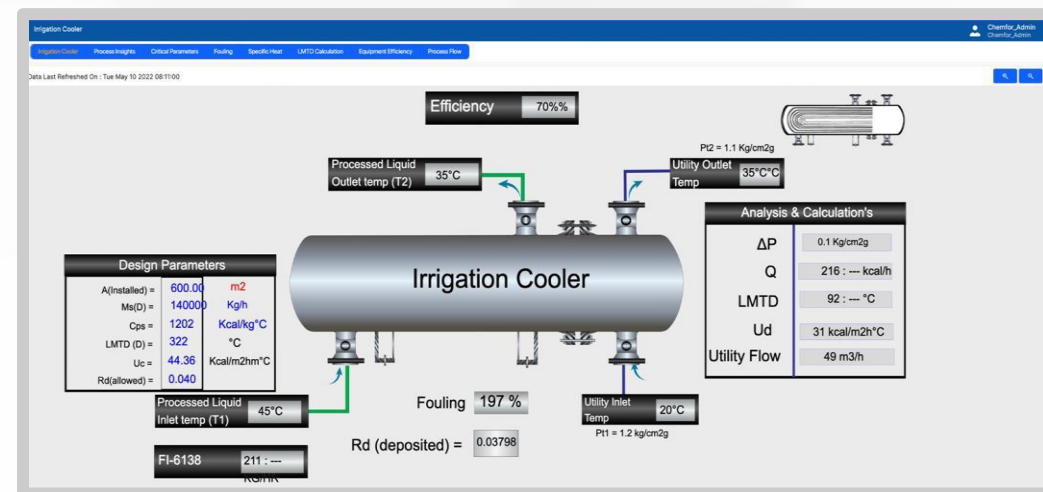
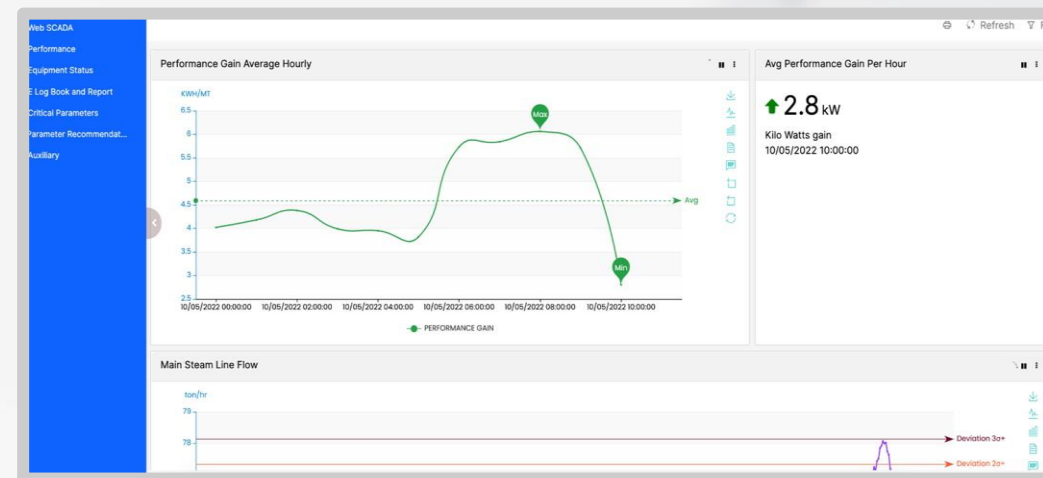
Asset Analytics

Solution Overview

Asset Analytics is a Digital Twin model driven asset analytics app for real time efficiency recommendations and predictive maintenance of critical assets

Key Features

- ✓ **Global Catalog of Asset Models:** Knowledge Lens has 100+ predefined asset models (templates) available that have the parameters, KPIs and business rules related to efficiency and maintenance of assets
- ✓ **Real Time Analytics on Asset operating profiles:** Data Explorer and real time dashboards are available for self-service analytics on all the critical parameters related to the asset, key KPIs like Asset Efficiency and also deviation between normal and anomalous operating profiles
- ✓ **Anomaly Detection and Alert generation:** AI model-based anomaly detection based on unsupervised learning as well as support for rule-based alert trigger
- ✓ **Integration with SAP/other enterprise maintenance systems:** Out of the box connectors/APIs to integrate with SAP/ERP or other systems that have maintenance history of the alert
- ✓ **Predictive Maintenance Model:** Suite of AI models for predicting Remaining Useful Life (RUL) based on historical fault data available for the asset
- ✓ **Mobile App for KPI & Ticket monitoring:** In case enterprise App is not available, there is feature support for monitoring KPIs and managing service history



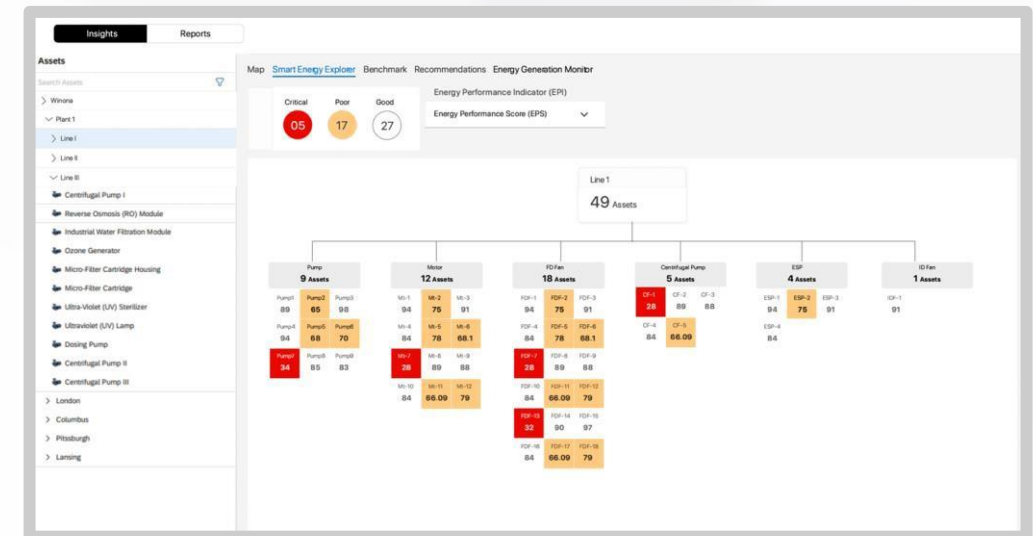
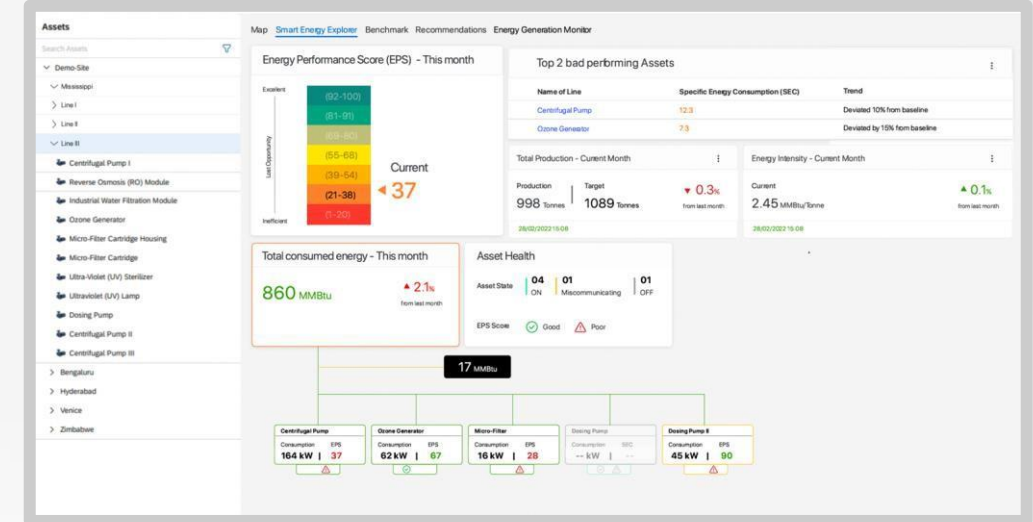
Energy Manager

Solution Overview

Energy Manager is an EMS app to monitor, analyze and optimize energy consumption across the various facilities of an industry

Key Features

- ✓ **Energy Star EPI Calculations:** Energy Performance Indicators are calculated automatically based on US EPA Standards
- ✓ **Loss Calculations and drill down:** Ability to identify various energy losses (e.g distribution loss, transformer loss, excess power consumption loss) etc at various levels of the site hierarchy
- ✓ **Critical Asset Energy Efficiency:** Provides detailed analysis of energy consumption by critical assets and provides recommendations on Energy Saving and optimization
- ✓ **Anomaly Detection:** AI Based anomaly detection engine to identify significant changes in energy consumption based on historical profiles of any asset
- ✓ **Benchmarks and Comparison:** Ability to compare energy performance of a site/plant/line/asset across similar levels of site hierarchy and improve the performance of poorly performing assets
- ✓ **Energy recommendations and ROI calculator:** Ability to calculate energy savings based on following the recommendations from the app and minimizing the energy losses



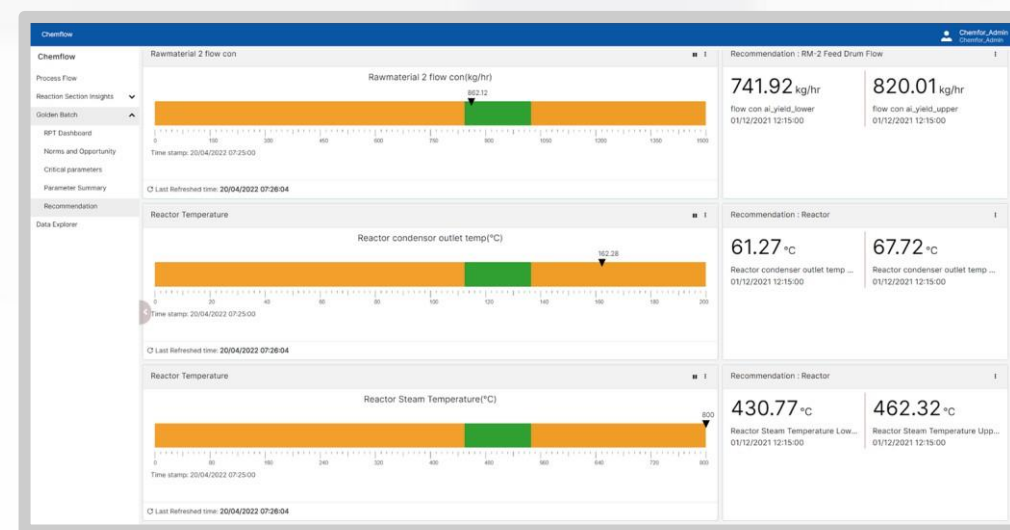
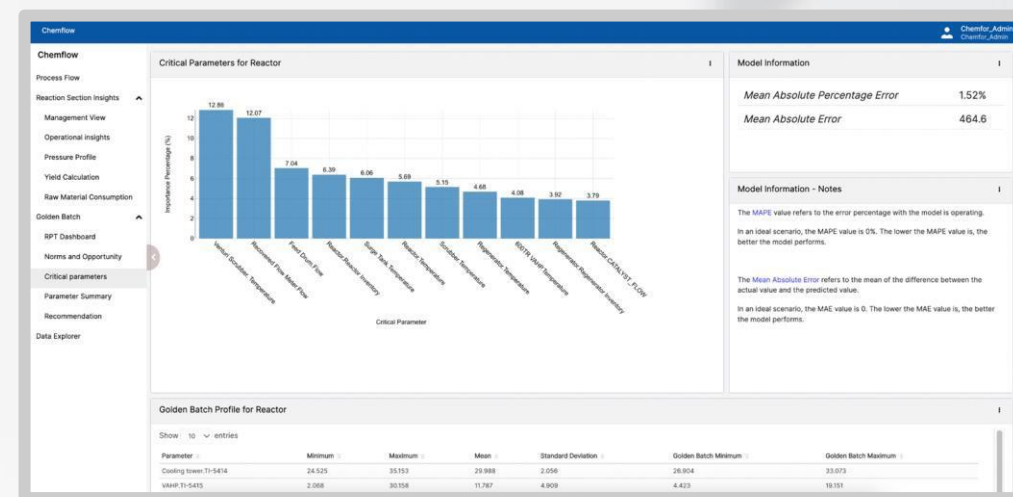
Yield Optimization

Solution Overview

Yield Optimization is an app to monitor, analyze and maximize production yield and efficiency for any continuous manufacturing process

Key Features

- ✓ **Operator-facing Recommendation App:** An intuitive and easy to use real-time app for operators that provides 'actions' they need to perform to achieve the optimum yield
- ✓ **AI Optimization Engine:** AI model that produces real time recommendation on the optimum operating range for the critical parameters and feeds the results into the Recommendation App
- ✓ **Auto-generation of Critical Parameters and importance:** Out of the hundreds of process parameters that impact the resulting yield, the app has a self-service UI that can be used to define the optimization targets and the list of process parameters. The AI model then generates the list of top 10 to 15 critical parameters along with their importance percentage
- ✓ **Yield Monitoring:** Real time visibility of process yields and yield trends for scenarios when the operator followed/ignored recommendations
- ✓ **Golden Batch Comparison:** Ability to define and compare Golden Batch parameters across shifts, business KPIs and operating days



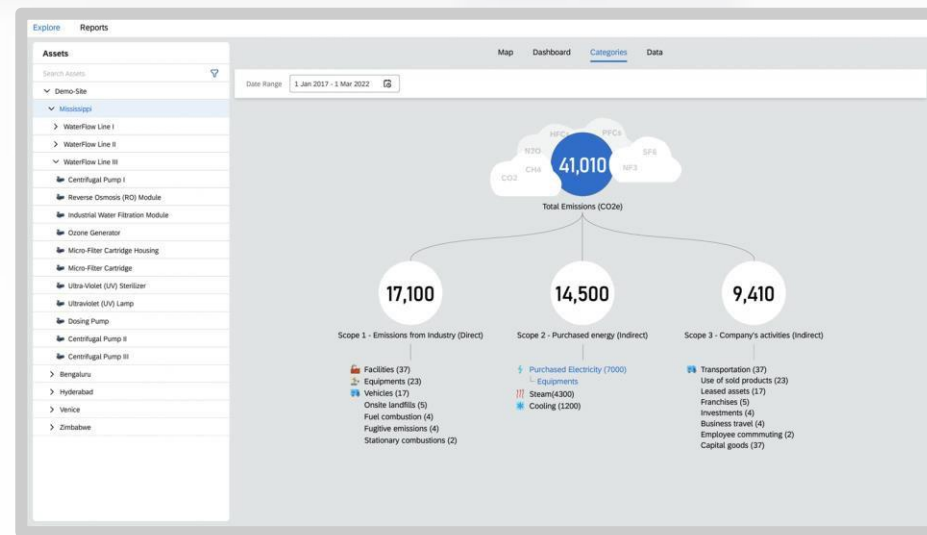
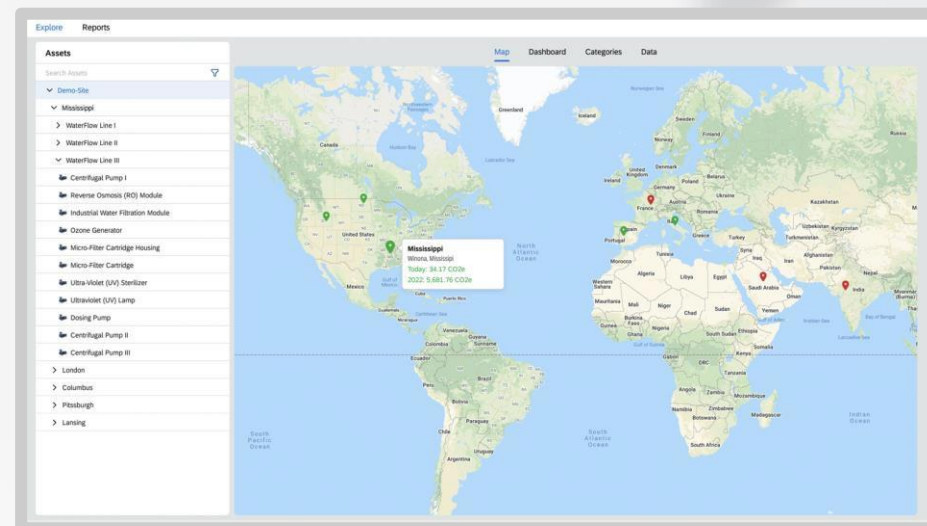
Sustainability

Solution Overview

Sustainability is an end-to-end product suite for Emissions Monitoring and Sustainability Analytics. It consists of 2 modules- the GLens DAHS and the Sustainability. With the GLens DAHS module, any industry can get MCERTS Certified Environment compliance along with NESHAP Emission Prediction. Sustainability module enables Carbon Accounting and Net Zero Reporting across Scope 1, Scope 2 and Scope 3 emissions. Companies can then generate pre-built reports compliant with different frameworks like GRI, SASB, TCFD etc.

Key Features

- ✓ **GLens DAHS:** MCERTS Certified Emission Compliance Reporting including Part 60 and Part 75 as per US EPA regulations as well as NESHAP Compliance along with prediction. Protocol support implemented across 3000+ customers for various sensors/analyzers relevant to GHG gases. Continuous and real time insights, reports, and dashboards on emission trends
- ✓ **Sustainability Reporting App:** Integration with GLens DAHS or third-party software for Emissions, Energy and Production Data. Support for manual data entry when IoT data is not available. Out of the box support for reporting compliant with industry standard reporting frameworks like GRI, SASB, TCFD



Customer Success Stories

Remote Monitoring - Case Study

Cement Manufacturer - a flagship company of \$1.73 billion Group

Problem Statement

- Unavailability of real-time monitoring of Critical Process Parameters
- Real-time Alerts to Operators was not available.
- Plant Corporate & Management Team didn't have a consolidated KPI Dashboards for their multi- plant operations under a single view.

What we provided

- Production Process Monitoring & KPI Dashboards
- Performance Indicators of Utilities
- Proactive control enabled to monitor Critical Process Parameters and trigger Alerts
- Reduced Operational Risk through triggering of Alerts for deviations, to be actioned by Plant team.

Business Outcome & ROI

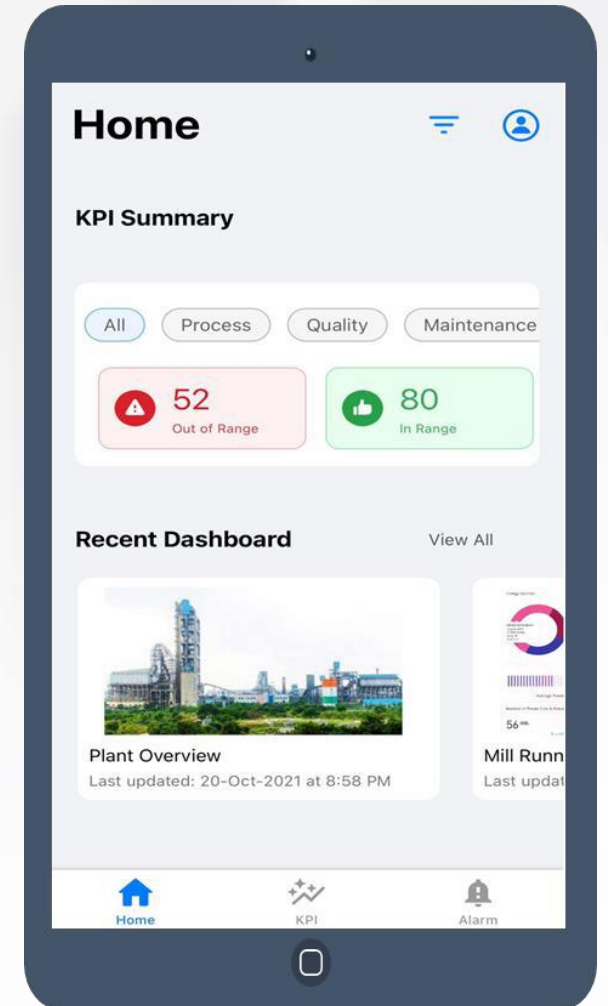
2X
IMPROVEMENT
IN
WORKER
PRODUCTIVITY

40%
DOWNTIME
REDUCTION
THROUGH
PROACTIVE
ALERTING FOR
KPI DEVIATIONS

100%
ELIMINATION
OF
MANUAL DATA
RECORDING BY
OPERATORS

Scalability

- Solution is repeatable and scaled to 14 Plants consisting of 25 Units consisting of Cement Integrated Units (CIU), Captive Power Plants (CPP), Waste Heat Recovery Plants (WHR)



Asset Analytics - Case Study

Textile, VSF & Chemicals Manufacturer - a flagship company of \$48.3 billion Group

Problem Statement

- Chiller Actual cooling capacity & efficiency not measured, monitored and analyzed on real-time basis
- Higher Energy consumption & Costs – Optimum loading conditions not being monitored in real-time
- Equipment Maintenance and Upkeep – Manual record of unplanned downtime

What we provided

- Chillers efficiency improvement
- Operational cost savings
- Improved levels of worker & environment safety
- Increase the life of Chillers through predictive maintenance



Business Outcome & ROI

5000
KWH PER
MONTH

COST PER
QUANTITY (KWH
PER TON) OF
CONSUMPTION

22%
OPTIMUM
SEQUENCING

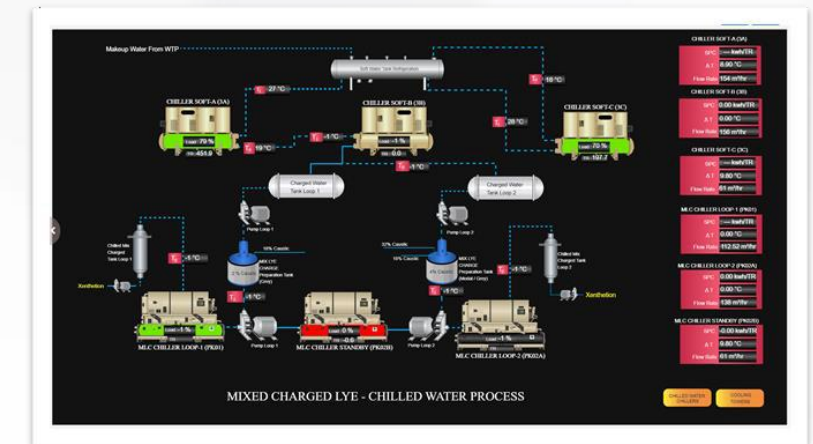
(ON/OFF,
SERIAL/PARALLEL) OF
CHILLERS – COST SAVINGS
PER PRODUCTION

20%
OPERATIONAL
COST

REDUCTIONS

Scalability

- Solution is repeatable and scaled to 20+ Critical Large assets within the large manufacturing facility.



Energy Manager - Case Study

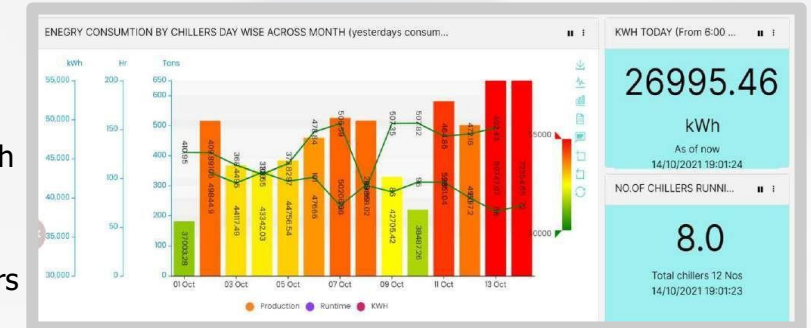
\$3.5B Textile/Carpet Manufacturer

Problem Statement

- No real time visibility on Energy consumption or various energy losses in plant
- Reactive mode for Energy Efficiency solutions done after yearly audit
- No real time alerts on potential problems in critical assets.

What we provided

- High Consumption of A/KW than Rated due to heavy loads than the Capacity
- Reduction of Higher Distribution Losses in Switch Gears/MDB /SSB, Transformer losses, Low PF machines.
- Automatic detection and notification to operators of conditions and Alarms
- Energy forecast of power consumption
- Workflow e-log book to track ,assign & closure of immediate recommendations.



Business Outcome & ROI

4.5%
SAVINGS
IN

ENERGY BILL

12%
REDUCTION
OF

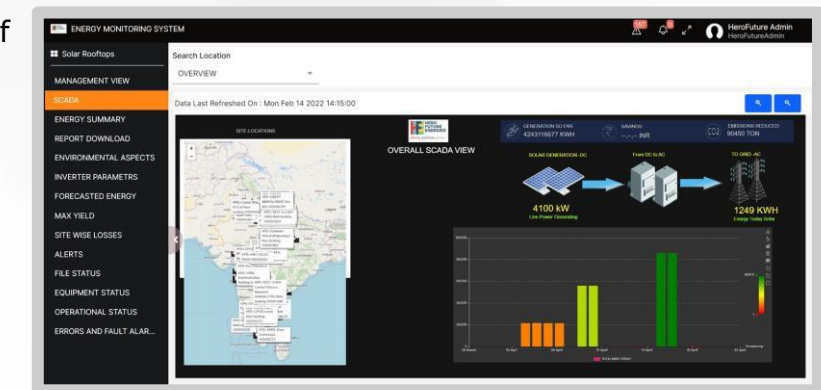
ENERGY LOST DUE TO
ABNORMALITIES &
INEFFICIENCIES OF
EQUIPMENT

04
MONTHS TO
BREAKEVEN

ON TECHNOLOGY
SPEND

Scalability

- Solution is repeatable and scaled to 15+ Plants



Yield Optimization - Case Study

\$1B Global Chemical Manufacturer

Problem Statement

- Unpredictable yield across various batches of manufacturing
- High dependency on skilled operators for efficient operations
- No visibility on operation actions

What we provided

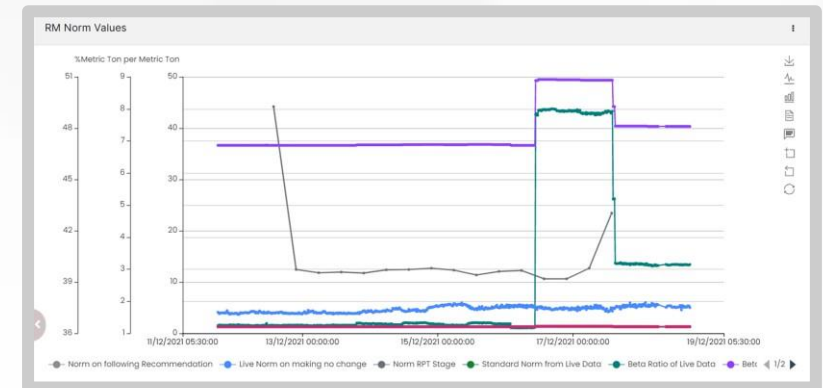
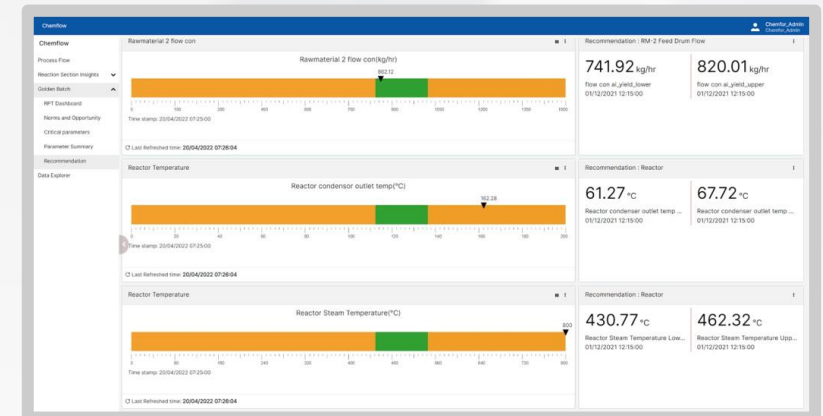
- Golden batch analytics to improve yield & optimize raw material usage
- Critical process parameters and their contribution to yield was identified
- Operational setpoints delivered to the shop floor operator as AI recommendations for Best Yield with Highest Profitability.

Business Outcome & ROI



Scalability

- Solution is repeatable and scaled to 15+ Plants



Sustainability - ESG monitoring Case Study

\$30B Global Metals & Minerals manufacturing conglomerate

Problem Statement

- Frequent incidents of emission exceedances - Intervention of regulatory bodies
- Non-uniform emission reporting - Difficulty in identifying source of non compliances
- Missed calibration and maintenance of Analyzers - Anomalies in critical parameter data

What we provided

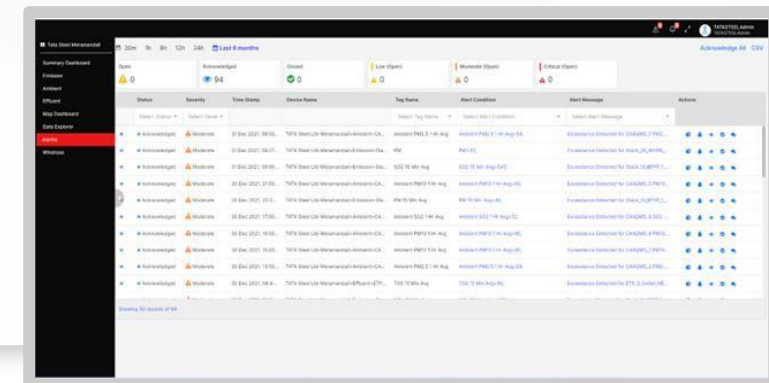
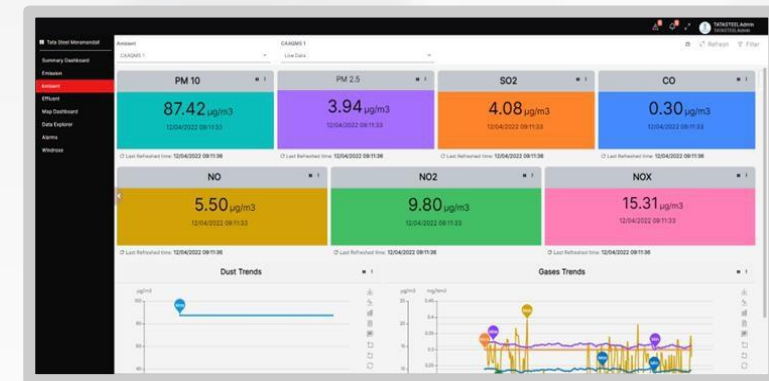
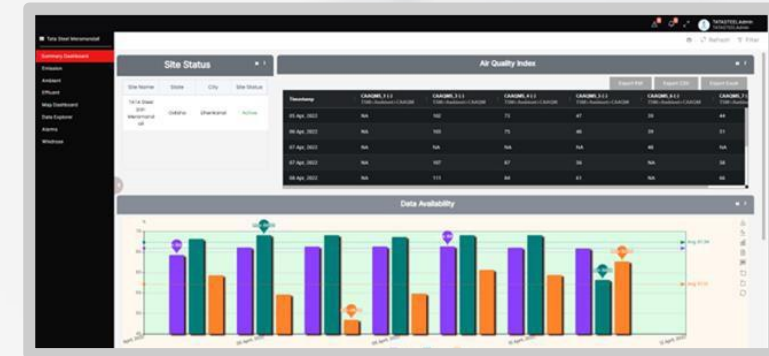
- Realtime dashboard for monitoring emissions at analyzer, station, plant and corporate level
- Fully compliant regulatory reports, custom reports as per company EHS guidelines
- Predictive/Proactive Alerts for possible exceedances of environmental parameters.

Business Outcome & ROI



Scalability

- Solution is repeatable and scaled to 5+ Plants
- Phase II - Integration for Carbon Footprint reporting covering Scope 1, Scope 2 and Scope 3 emissions



Sustainability - NESHAP Prediction Case Study

\$23B Sustainable Global Building Solutions

Problem Statement

- On-Prem Solutions not able to scale or standardize
- Local DAHS System not able to Forecast NESHAP Compliance
- Manual and inaccurate scenario modeling tools using local excel sheets.

What we provided

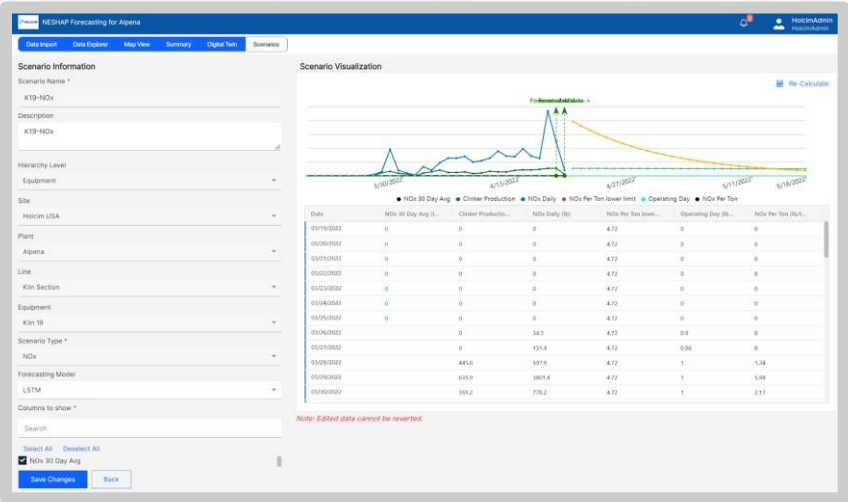
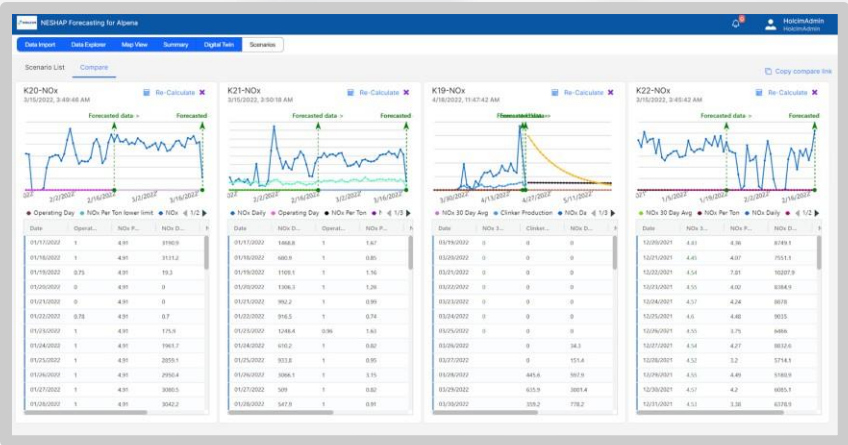
- Enabled Cloud DAHS for visibility across plants
- Standardized accurate NESHAP Compliance Forecast on real time data from DAHS
- Scenario modeling tool on IoT Data for performing what-if analysis using Digital Twin

Business Outcome & ROI



Scalability

- Solution is repeatable and scaled to 20+ Critical Large assets within the large manufacturing facility.



Thank You



**Rockwell
Automation**

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